

LEGALEASE

3. Project Design Phase

LegalEase – System Architecture and Design

PROJECT TEAM

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1. Design Goal

Prepared using the supplied Periyar University-style reference format and the supplied LegalEase project document. The reference structure covers project title/description, use cases, architecture, prerequisites, objectives, workflow, milestones, testing, deployment, results, limitations, future enhancements and appendix. The LegalEase source describes an AI-powered legal document generator using Streamlit, FastAPI, Gemini integration and TXT/DOCX/PDF export.

2. High-Level Architecture

User → Streamlit Frontend → FastAPI Backend → Gemini AI Core → Generated Text → Formatting Modules → TXT/DOCX/PDF.

3. Component Design

- Frontend: Streamlit
- Backend: FastAPI and routes
- AI Core: GeminiDocumentGenerator
- Formatting: DOCX/PDF/HTML utilities
- Configuration: .env

4. User Interface Design

The interface accepts document type, parties, terms and effective date, then provides generation, editing and download actions.

5. Data Flow

- User input
- Frontend request
- Backend validation
- Structured AI prompt
- Gemini response
- Preview
- User editing
- File export

6. Formatting Design

DOCX output can contain logo, headings, paragraph formatting, terms table and footer. PDF output can contain logo, headings, bullet-style terms and footer.

7. Design Principles

- Modularity

- Simple user experience
- Separation of concerns
- Editable output
- Secure configuration
- Future deployment readiness